X-ray photoelectron spectroscopy (XPS) of Quinacridone (QA) on the Ag(100) surface

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Motivation

- QA shows 1D molecular chains due to intermolecular hydrogen bonds[1–4]
- Previous studies revealed 2 distinct phases of QA on Ag(100)
- **But:** Molecular interactions & energetics of phase transition are not fully understood

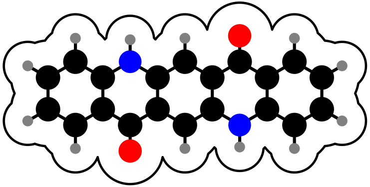


Figure: Ball-and-stick model of QA: C: black, H: grey, N: blue, O: red.

Research Question

How do intermolecular hydrogen bonds & adsorbate-substrate interactions change during phase transitions?

Known α-Phase of QA on Ag(100)

α-Phase:

- Forms at 300 K
- point-on-line (POL) structure
- Homochiral 1D chains
- Proposed 2 hydrogen bonds per molecule
- Metastable phase

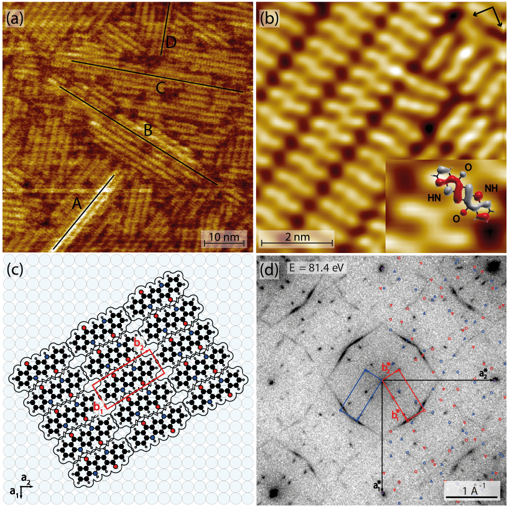


Figure: α-phase of QA on Ag(100).[1]

Known β-Phase of QA on Ag(100)

β-Phase:

- Forms after heating to 500 K
- Commensurate structure
- 2D network with dimers
- Proposed 1.5 hydrogen bonds per molecule
- Thermodynamically stable

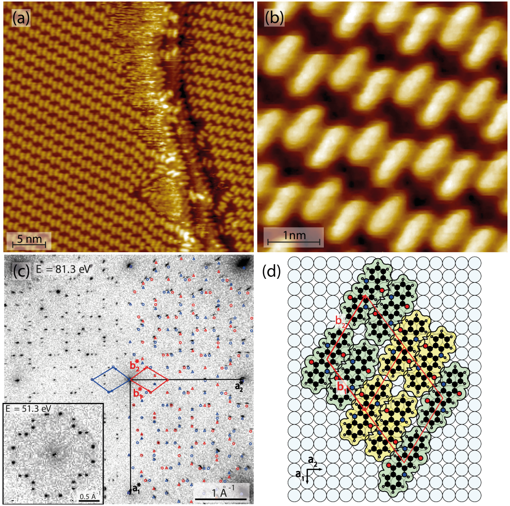


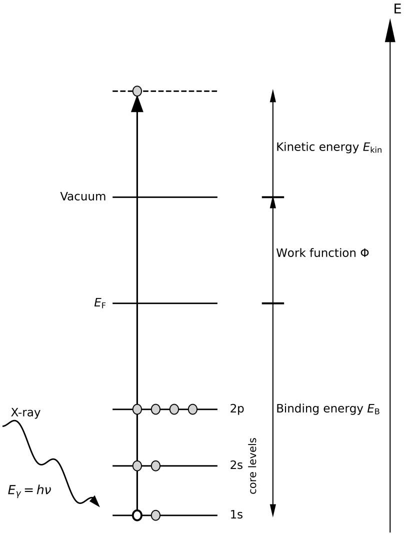
Figure: β-phase of QA on Ag(100).[1]

X-ray photoelectron spectroscopy

$$E_{\text{kin}} = h\nu - E_B - \phi_{\text{sample}}$$

Key Features:

- Surface-sensitive ($\sim$ nm depth)
- Element-specific core levels
- Binding energy reflects local environment



Experimental Setup

Location: Diamond Light Source (I09 beamline), UK

- UHV chamber ($p \approx 10^{-10}$ mbar)
- Synchrotron radiation source
- Variable photon energy

Figure: Diamond Light Source, UK.[5]

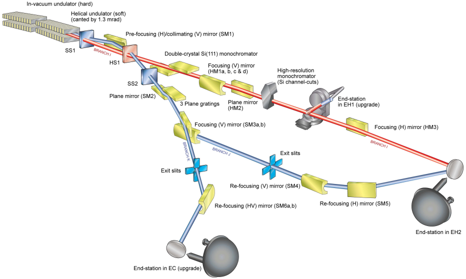


Figure: I09 endstation schematic.[5]

C1s XPS Spectra

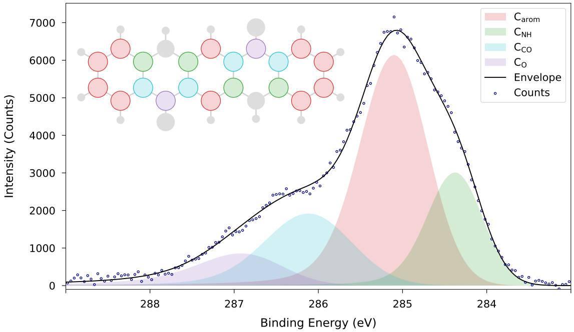


Figure: C1s spectrum of α-phase.

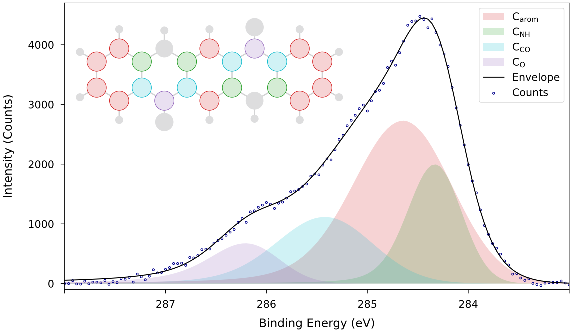


Figure: C1s spectrum of β-phase.

N1s XPS Spectra

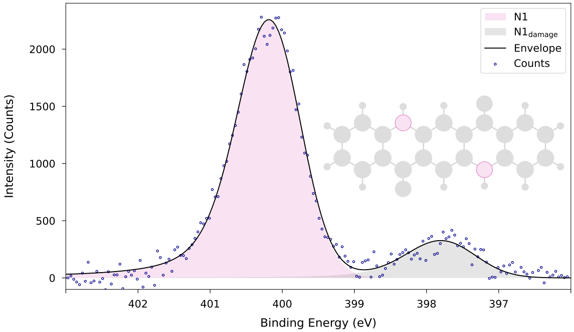


Figure: N1s spectrum of α-phase.

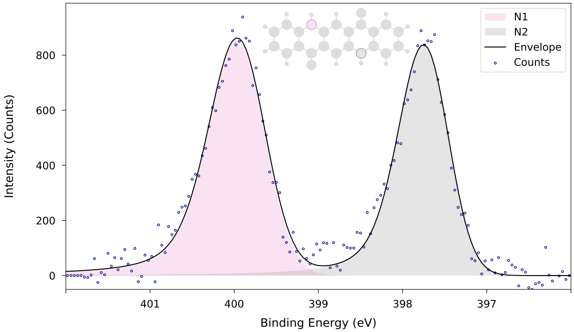


Figure: N1s spectrum of β-phase.

O1s XPS Spectra

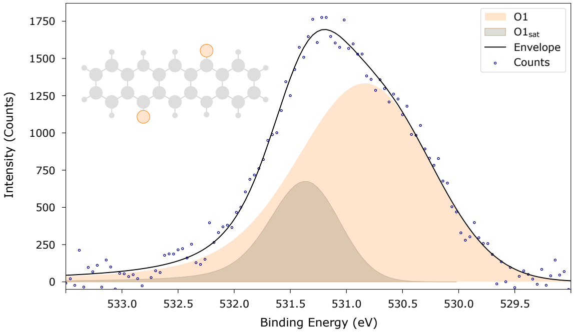


Figure: O1s spectrum of α-phase.

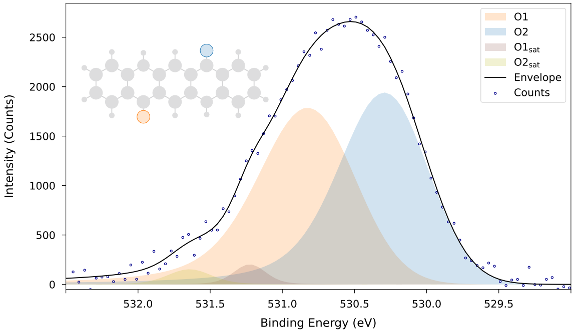


Figure: O1s spectrum of β-phase.

Binding Energy Changes

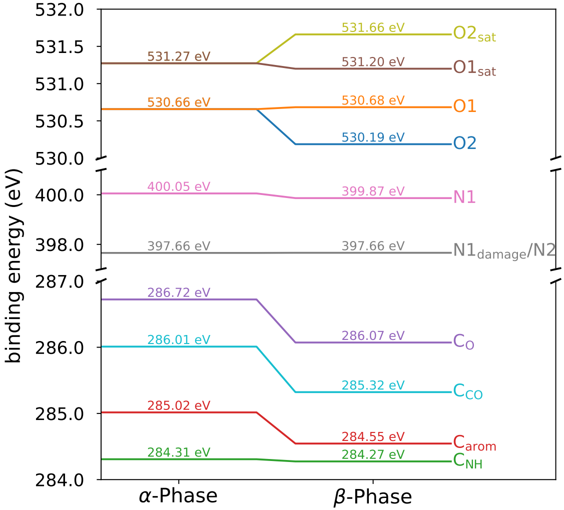


Figure: Binding energy shifts from α- to β-phase.

- **General trend:** Lower BEs in β-phase $\rightarrow$ stronger surface interaction
- **C1s:** Uniform shifts except for C_{NH}
- **N1s:** Large 2.2 eV shift for N2 component
- **O1s:** 0.5 eV difference between O1 and O2

N1s Shift:

- 2.2 eV shift matches literature for deprotonated amine group[6–8]
- Similar to beam damage in α-phase

Supporting Evidence:

- O1s shows 2 distinct environments
- One O maintains H-bonding (O1)
- Other O loses H-bonding partner (O2)

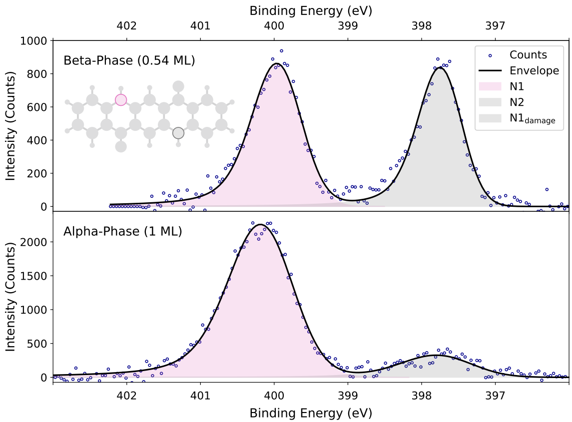


Figure: N1s XPS spectra for α- & β-phase.

Key Findings

α-Phase

- Single peaks for N and O atoms $\rightarrow$ equivalent environments
- Confirms 2 hydrogen bonds per molecule

β-Phase

- 2 distinct N & 2 distinct O environments (1:1 area ratio)
- Large N1s shift suggests deprotonation of 1 amine group
- Evidence for only 1 hydrogen bond per molecule

New Interpretation

Phase transition involves deprotonation of 1 amine group per molecule!

Structure Models: α- vs. β-Phase

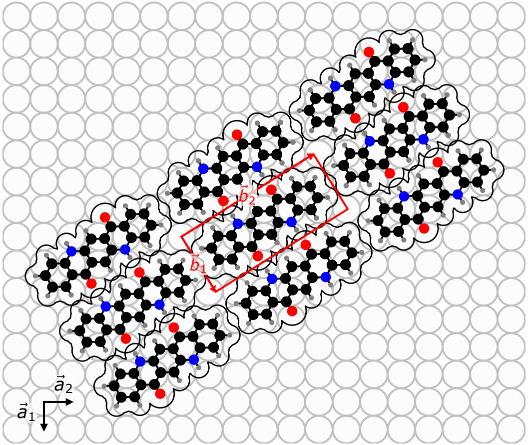


Figure: α-phase: 2 H-bonds per molecule, all N and O atoms equivalent

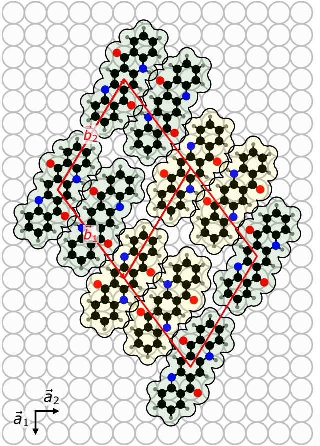


Figure: β-phase: 1 H-bond per molecule, one N deprotonated

Summary of Main Results

1. **Confirmed α-phase structure:**
   - 2 hydrogen bonds per molecule
   - All N & all O atoms have equivalent environments
   - Consistent with existing structure model
2. **Revised β-phase understanding:**
   - 1 hydrogen bond per molecule (not 1.5)
   - 1 deprotonated amine group per molecule, 2 distinct N & 2 distinct O environments
   - Requires new structure model
3. **Phase transition:**
   - Deprotonation of amine group
   - Stronger molecule-surface interactions in β-phase

Key Impact

New insights into molecular interactions and phase behavior of QA on the Ag(100) surface, including evidence for deprotonation during phase transition.

Future Perspectives

Follow-up Studies

- **Normal incidence X-ray standing wave (NIXSW):** Determine exact adsorption heights and sites
- **Mass spectrometry:** Detect H_2 evolution during phase transition
- **Other derivatives:** Study similar systems, e.g. FQA

References I

[1] Hans Niklas Humberg. "One-Dimensional Aggregates of the Organic Dye Quinacridone on Metallic and Dielectric Surfaces". Bonn: Rheinische Friedrich-Wilhelms-Universität Bonn, 2024.

[2] Thorsten Wagner et al. "Quinacridone on Ag(111): Hydrogen Bonding versus Chirality". In: *The Journal of Physical Chemistry C* 118.20 (May 22, 2014), pp. 10911–10920.

[3] Frank Trixler et al. "Supramolecular Self-Assembly Initiated by Solid–Solid Wetting". In: *Chemistry – A European Journal* 13.27 (2007), pp. 7785–7790.

[4] Alexander Richard Eberle. "Growing Low-Dimensional Supramolecular Crystals via Organic Solid-Solid Wetting Deposition". Ludwig-Maximilians-Universität München, Feb. 2019.

[5] Diamond. *Diamond Light Source: I09*. June 2025. URL: `https://www.diamond.ac.uk/Instruments/Structures-and-Surfaces/I09.html`.

[6] G. Ruano et al. "Deprotonation of the Amine Group of Glyphosate Studied by XPS and DFT". In: *Applied Surface Science* 567 (Nov. 30, 2021), p. 150753.

[7] E.T. Kang, K.G. Neoh, and K.L. Tan. "XPS Studies of Charge-Transfer Interactions in Some Pyridine: Organic-Acceptor Complexes". In: *Molecular Physics* 70.6 (Aug. 20, 1990), pp. 1057–1064.

References II

[8] Muzamir M. Mahat et al. “Elucidating the Deprotonation of Polyaniline Films by X-ray Photoelectron Spectroscopy”. In: *Journal of Materials Chemistry C* 3.27 (2015), pp. 7180–7186.

[9] Raphael Schlesinger. “Theory of Experimental Methods”. In: *Energy-Level Control at Hybrid Inorganic/Organic Semiconductor Interfaces*. Ed. by Raphael Schlesinger. Cham: Springer International Publishing, 2017, pp. 47–76.

[10] Oliver Bauer. “Surface Bonding of a Functionalized Aromatic Molecule: Adsorption Configurations of PTCDA on Coinage Metal Surfaces”. Rheinische Friedrich-Wilhelms-Universität Bonn, 2014.

[11] J. Brunet et al. “Physical and Chemical Characterizations of Nanometric Indigo Layers as Efficient Ozone Filter for Gas Sensor Devices”. In: *Thin Solid Films* 520.3 (Nov. 2011), pp. 971–977.

[12] Leander Michels et al. “Electronic and Structural Properties of the Natural Dyes Curcumin, Bixin and Indigo”. In: *RSC Advances* 11.23 (2021), pp. 14169–14177.

[13] Hongxiang Xu et al. “On-Surface Isomerization of Indigo within 1D Coordination Polymers”. In: *Angewandte Chemie* 136.15 (2024), e202319162.

Thank you for your attention!

Questions?

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Quinacridone Properties

Chemical Properties:

- Formula: $C_{20}H_{12}N_2O_2$
- Prochiral organic molecule
- Highly conjugated π-system
- Symmetry group: C_{2h}

Applications:

- Pigment in industry (Violet 19)
- Well-defined surface structures
- Organic electronics components

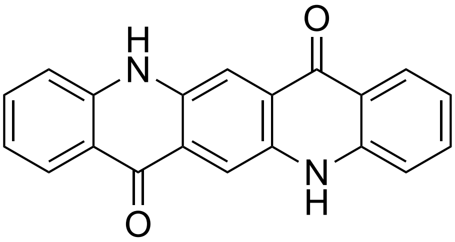

Figure: Structure of QA.

Multielectron processes in XPS

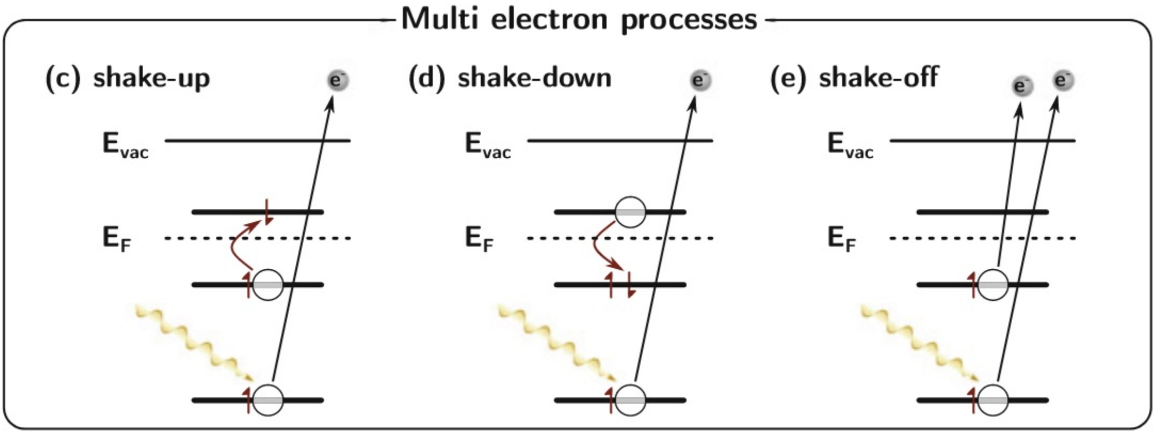

Figure: Multielectron processes in XPS. Taken from [9].

Literature Comparison

Table: Comparison of experimental results with literature values.

System	C_{arom} BE (eV)	N1s BE (eV)	O1s BE (eV)
QA/α-phase	285.02	400.05	530.66
QA/β-phase	284.55	399.87/397.66	530.68/530.19
PTCDA[10]	284.91	-	531.93
Indigo[11–13]	285.5	399.9	530.2

- Good agreement with similar aromatic systems
- Large N1s shift consistent with deprotonation studies
- O1s values typical for carbonyl groups
- Phase transition shifts indicate stronger surface coupling

LEED Patterns

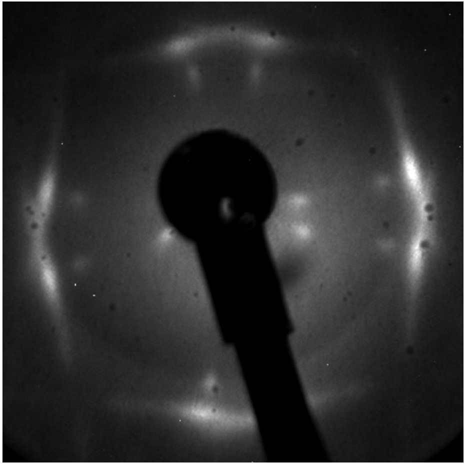

Figure: Recorded LEED pattern of α-phase at DLS.

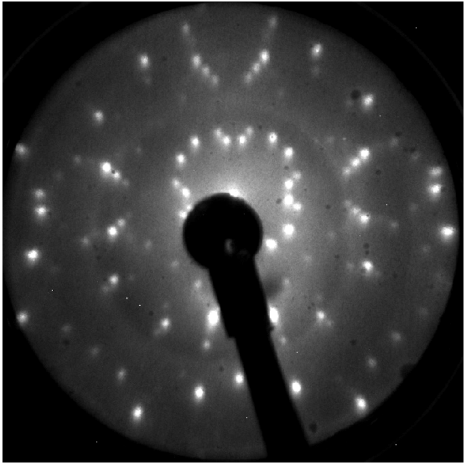

Figure: Recorded LEED pattern of β-phase at DLS.

Superstructure of QA on Ag(100)

- POL structure for α-phase[1]
  $\rightarrow$ every point of superstructure falls on substrate lattice line in [10] direction

$$\mathbf{M}_\alpha = \begin{pmatrix} 2 & 1.25 \\ -3 & 4.80 \end{pmatrix}$$

- $b_1 = (6.8 \pm 0.1)$ Å
- $b_2 = (16.4 \pm 0.1)$ Å
- $\alpha = 90^\circ \pm 1^\circ$

- Commensurate structure for β-phase[1]

$$\mathbf{M}_\beta = \begin{pmatrix} 4 & 3 \\ -5 & 3 \end{pmatrix}$$

- $b_1 = (14.4 \pm 0.1)$ Å
- $b_2 = (16.8 \pm 0.1)$ Å
- $\alpha = 112^\circ \pm 1^\circ$

C1s spectra

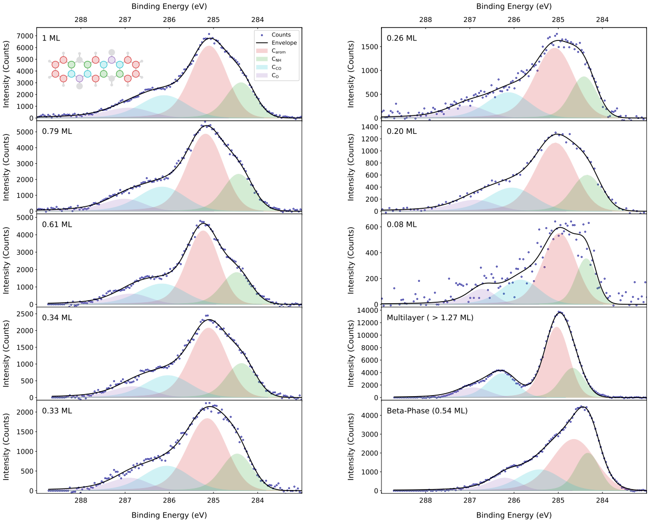

Figure: Overview of C1s XPS spectra for QA on Ag(100).

N1s spectra

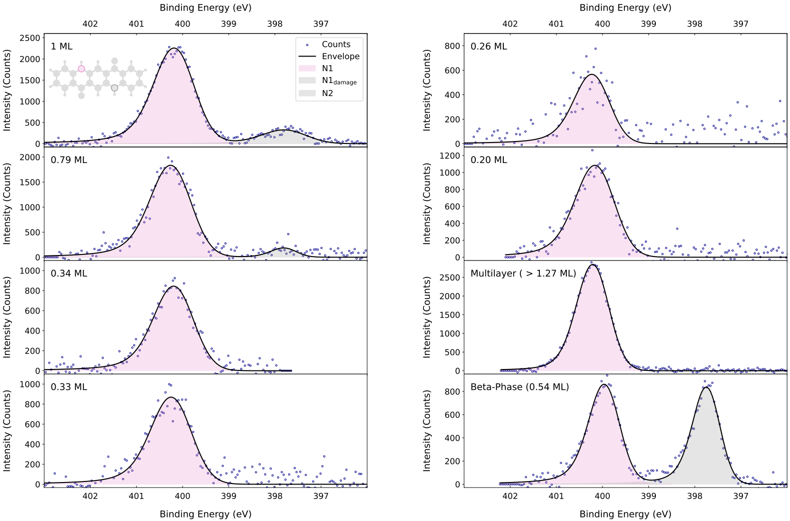

Figure: Overview of N1s XPS spectra for QA on Ag(100).

O1s spectra

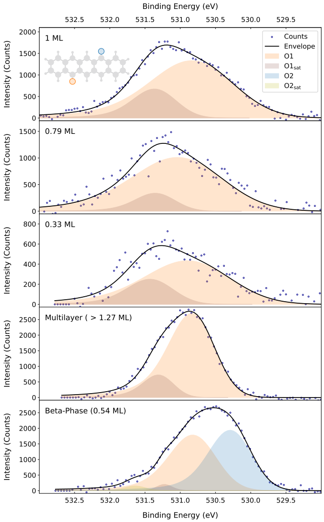

Figure: Overview of O1s XPS spectra for QA on Ag(100).

XPS fitting parameter

peak	line shape	area ratio
C_{arom}	LA(1,8,800)	10
C_{NH}	LA(1,8,800)	4
C_{CO}	LA(1,8,800)	4
C_{O}	LA(1,8,200)	2
O1	LA(1,8,400)	1 (for β-phase)
$O1_{sat}$	LA(1,1,400)	-
O2	LA(1,8,400)	1 (for β-phase)
$O2_{sat}$	LA(1,1,400)	-
N1	LA(1,8,400)	1 (for β-phase)
$N1_{damage}$	LA(1,8,400)	-
N2	LA(1,8,400)	1 (for β-phase)

Parameters of XPS fits

Phase	Core-level	Component	BE (eV)	FWHM (eV)	Area
α	C1s	C_{arom}	285.02	0.94	10
		C_{NH}	284.31	0.76	4
		C_{CO}	286.01	1.20	4
		C_{O}	286.72	1.15	2
	N1s	N1	400.05	1.02	-
		$N1_{damage}$	397.66	1.16	-
	O1s	O1	530.66	1.36	-
		$O1_{sat}$	531.27	0.71	-
β	C1s	C_{arom}	284.55	1.13	10
		C_{NH}	284.28	0.62	4
		C_{CO}	285.32	1.11	4
		C_{O}	286.07	0.79	2
	N1s	N1	399.87	1.20	1
		N2	397.66	1.20	1
	O1s	O1	530.68	0.71	1
		$O1_{sat}$	531.20	0.45	-
		O2	530.19	0.68	1
		$O2_{sat}$	531.66	0.36	-